


Fall 2026

# WEEK 12 STUDY GUIDE

## The Big Picture

We continue with inference for the unknown *p* of a coin, from a Bayesian perspective. Then we move to least squares estimation.

- The beta family is a rich class with which to describe our *prior* opinions about p; it then turns out that the same family describes our *posterior* opinion which is the prior updated based on the observed heads and tails.
- If you have the scatter diagram of simulated (X,Y) pairs, then Data 8 ideas say that given X, the best predictor of Y is the "center of the vertical strip at X." Formally, "best" means "least squares," and the "center of the vertical strip at X" is the conditional expectation of Y given X.
- The error in this estimate, given X, is the conditional SD of Y given X.

## Week At a Glance

| Mon 11/9 | Tue 11/10 | Wed 11/11 | Thu 11/12 | Fri 11/13 |
|---|---|---|---|---|
| Office Hours 10AM to 3PM | Lecture | Holiday | Lecture | Mega Section |
| **HW 11 Due at 5PM**<br>HW 12 (Due 5PM Mon 11/16) | | | | HW 12 Party 1PM to 4PM |
| **Lab 6A Due at 5 PM**<br>Lab 6B (Due 5PM Mon 11/16) | | | Lab 6B Party 4 PM to 6 PM | |
| | | | Work through Sections 22.1, 22.2. | Work through Sections 22.1, 22.2, 22.3, and Example 22.4.1 |

## Reading, Practice, and Class Meetings

| Book | Topic | Lectures: Professor | Sections: TAs | Optional Additional Practice |
|---|---|---|---|---|
| Ch 21 | **Inference for the *p* of a random coin**<br>- 21.1 picks up from 20.3, with a general beta prior instead of uniform<br>- 21.2 is about the unconditional distribution of the number of heads, which is called beta-binomial<br>- 21.3 is omitted this term | Tuesday 11/10<br>- Inference for the random p of a coin<br>- Conjugate priors; prediction<br>- Relations between the beta and the binomial, including the beta-binomial distribution | Wednesday 11/6<br>No sections (holiday!) | Ch 21<br>- All exercises not completed in homework |
| Ch 22 | **An approach to prediction**<br>- 22.1 develops the main reason why conditional expectation is important for prediction<br>- 22.2 shows that conditional expectation is a least squares predictor, and defines the error in the estimate | Thursday 11/12<br>- The random variable equivalent of “dropping a perpendicular”<br>- Least squares prediction, and a new variance | Friday 11/13<br>- Ch 22 Ex 6, 1, 7 | Ch 22<br>- Ex 2 |